

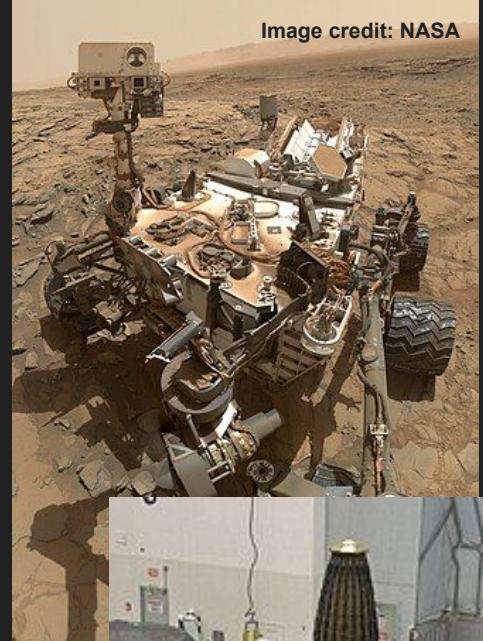
Lesson 5: The Space Shuttle

**NASA and Space Exploration:
A Journey through the Solar System**

Image credit: NASA

Recap from Last Week

- Humans have sent rovers and probes to all the planets in the Solar System
- What are the most important parts of a probe?
 - Solar panels, antenna, payload (tools inside probe), thrusters
- What are the most important parts of a rover?
 - Wheels, battery/solar panels, camera, drills, science instruments, antenna
- Where have we sent most of the rovers and probes?
 - Venus, Mars, and the Sun



Recap from Last Week

- Where have we only sent 1 probe?
 - Uranus, Neptune, and Pluto
- Some big discoveries from probes?
 - Venus: we took pictures and found wind and mountains
- Mars: dried lake and rivers (it used to have liquid water!), mountains and craters
- Jupiter: its poles are very different than its equator
- Future missions will look for life on the moons of Jupiter and Saturn, one even uses a helicopter

Image credit: NASA

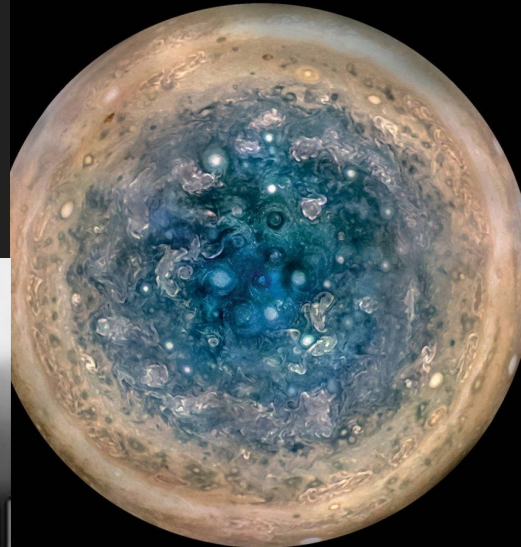
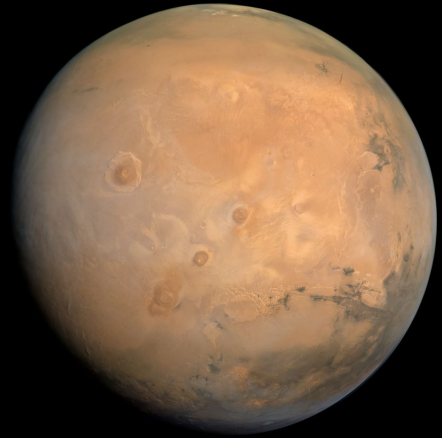


Image credit: Venera

It's the Curiosity Rover's Birthday!!



Last Week's homework

- Did anyone spot the ISS?

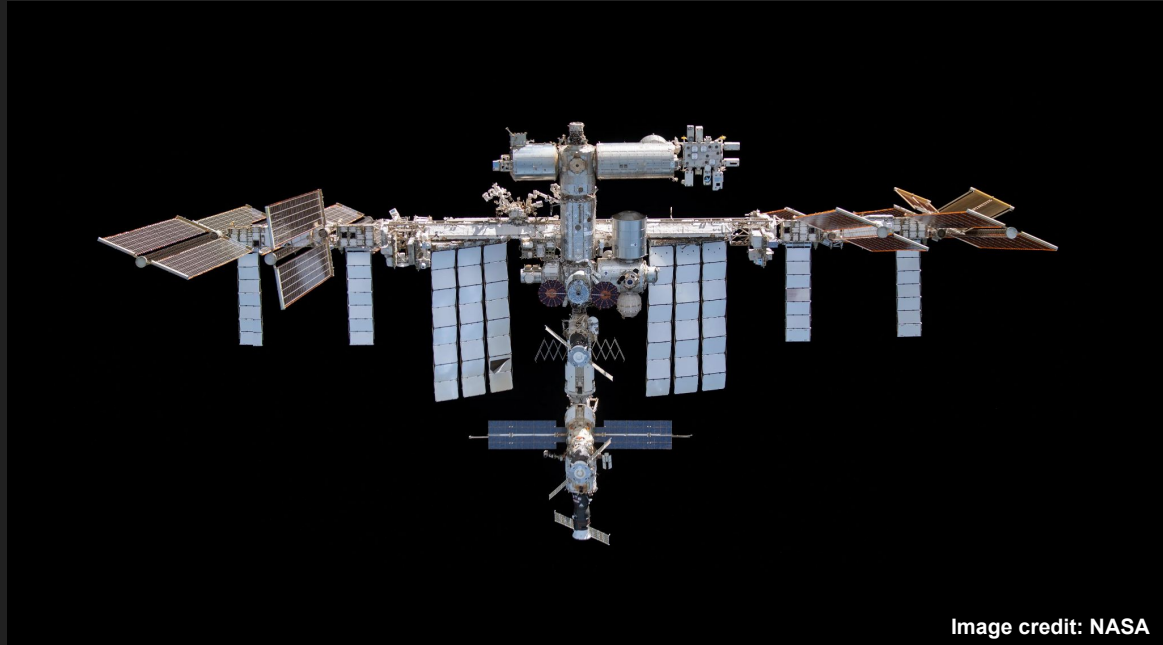


Image credit: NASA

Artemis and the Future of Space Exploration

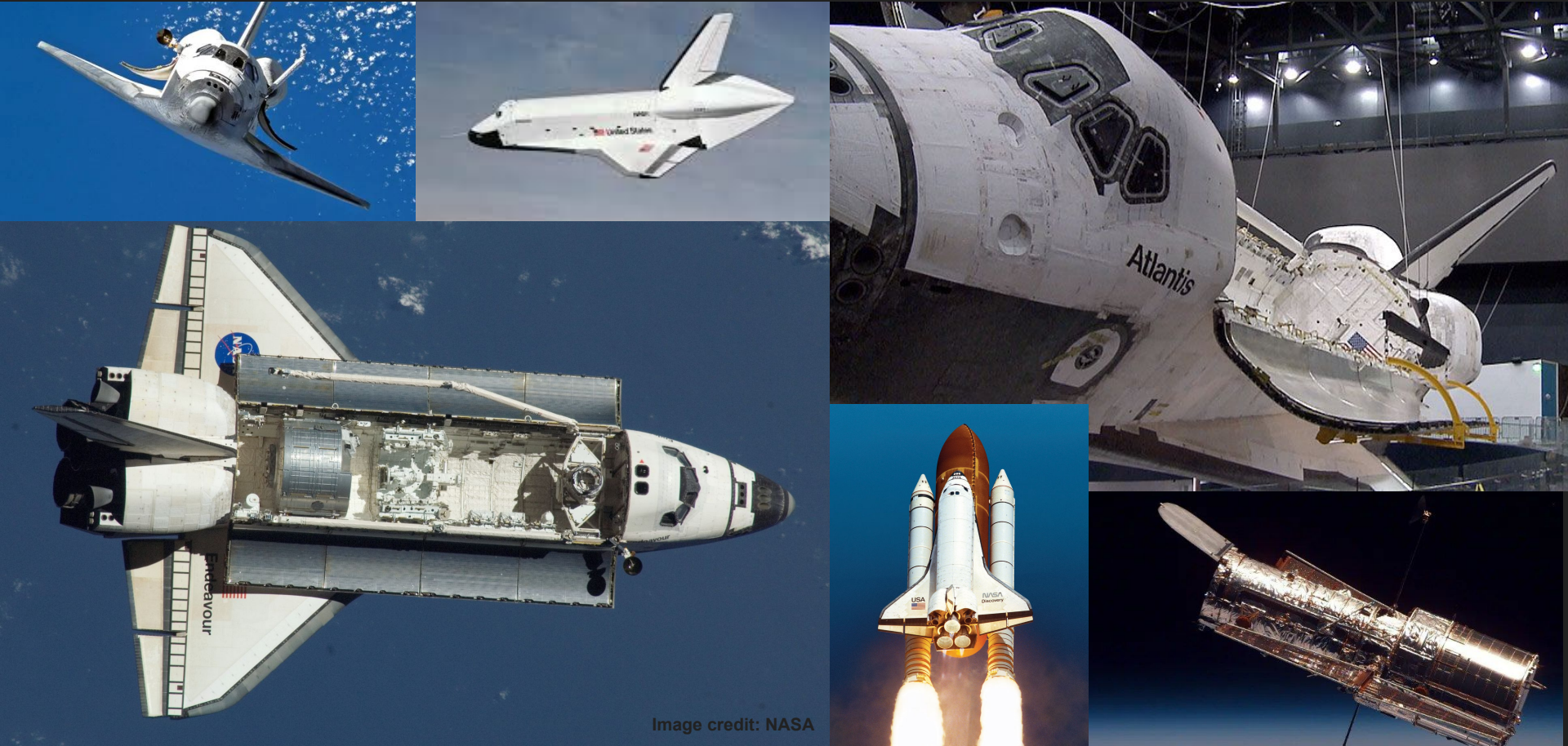


Image credit: NASA

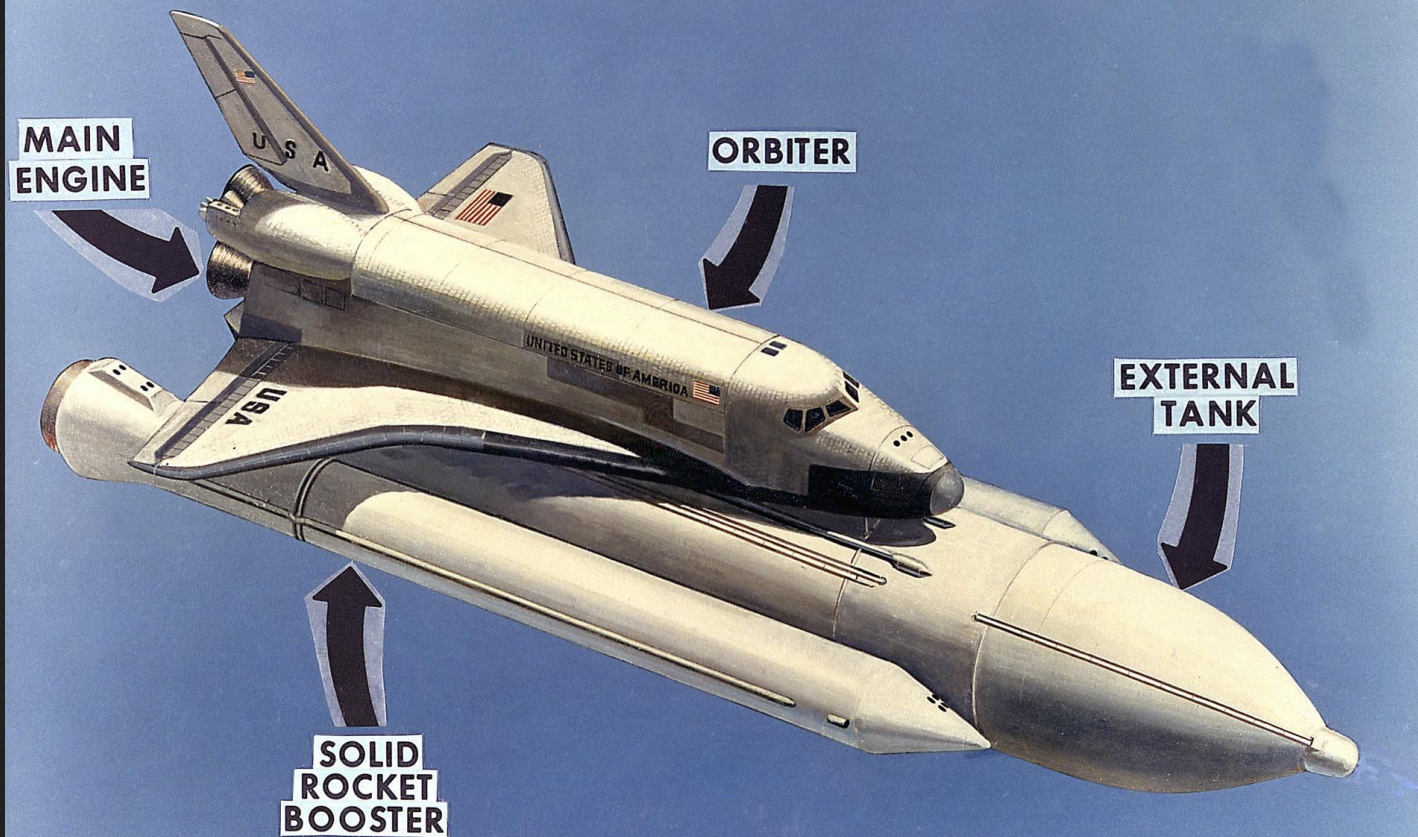
The Space Shuttle Program

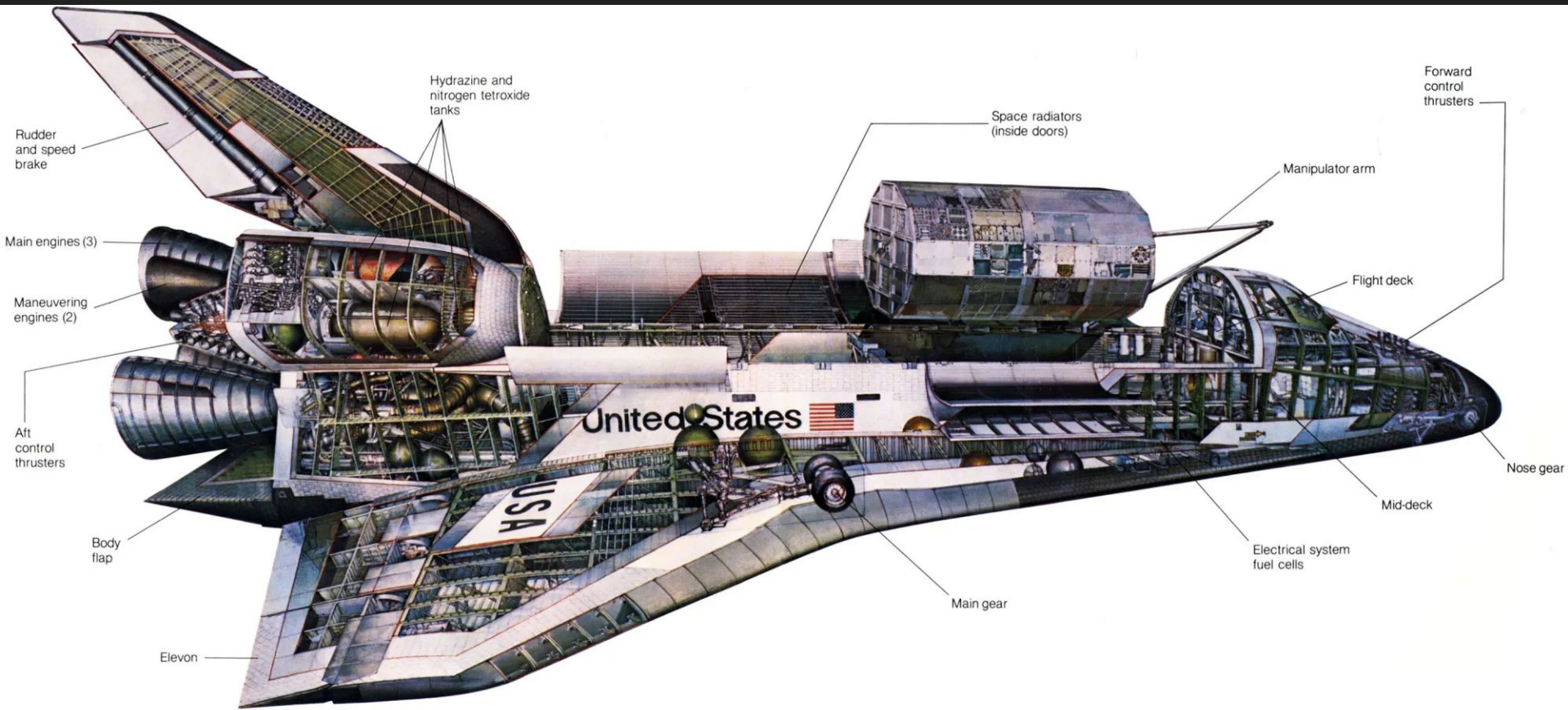
- Each time we flew to the Moon or into space during the Mercury, Gemini, and Apollo missions, the ships could not be used again
- Even today, each ship (Soyuz) sent to the International Space Station is built specifically for one mission
- The USA wanted to make a reusable spaceship that could be used for many missions
- This would make each flight cheaper and faster to go into space
- This started the Space Shuttle program in 1981

Quiz

- Where are the landing vehicles used to land on the Moon?
 - Still on the Moon!
- Where are the rovers that astronauts drove on the Moon? (Online only answer)
 - Still on the Moon!
- Where are the rovers that NASA sent to Mars? (Online only answer)
 - Still on Mars!
- Why are all the vehicles, capsules, and ships so expensive?
 - They are highly unique, specialized, and you have to make a new one each time you fly
- They are not made like your car that is mass produced in a factory

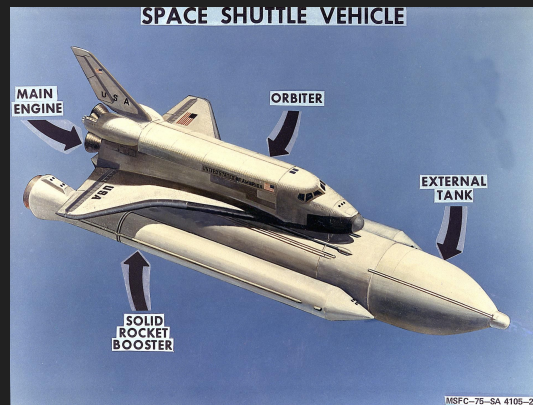
SPACE SHUTTLE VEHICLE





Launching the Space Shuttle

- The space shuttle orbiter (part that looks like a plane) would launch connected to two solid boosters and a gas tank
- The orbiter would separate from the boosters and the gas tank
- The orbiter had its own rockets that could control its motion in space and its landing
- The boosters would land in the ocean using a parachute and be reused
- The gas tank was the only part not reused
- The orbiter landed like a plane







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AP





COLUMBIA



CHALLENGER



DISCOVERY

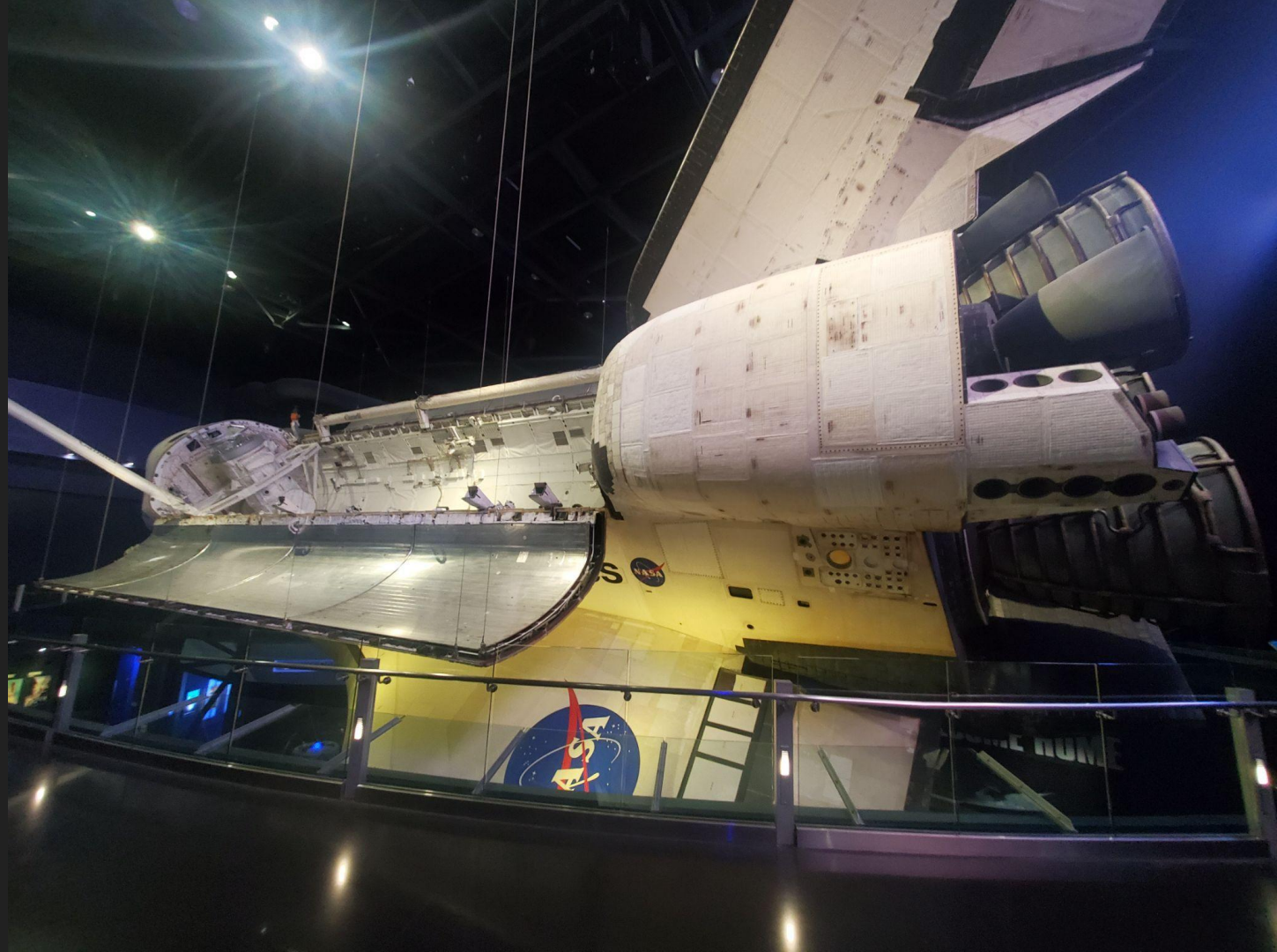


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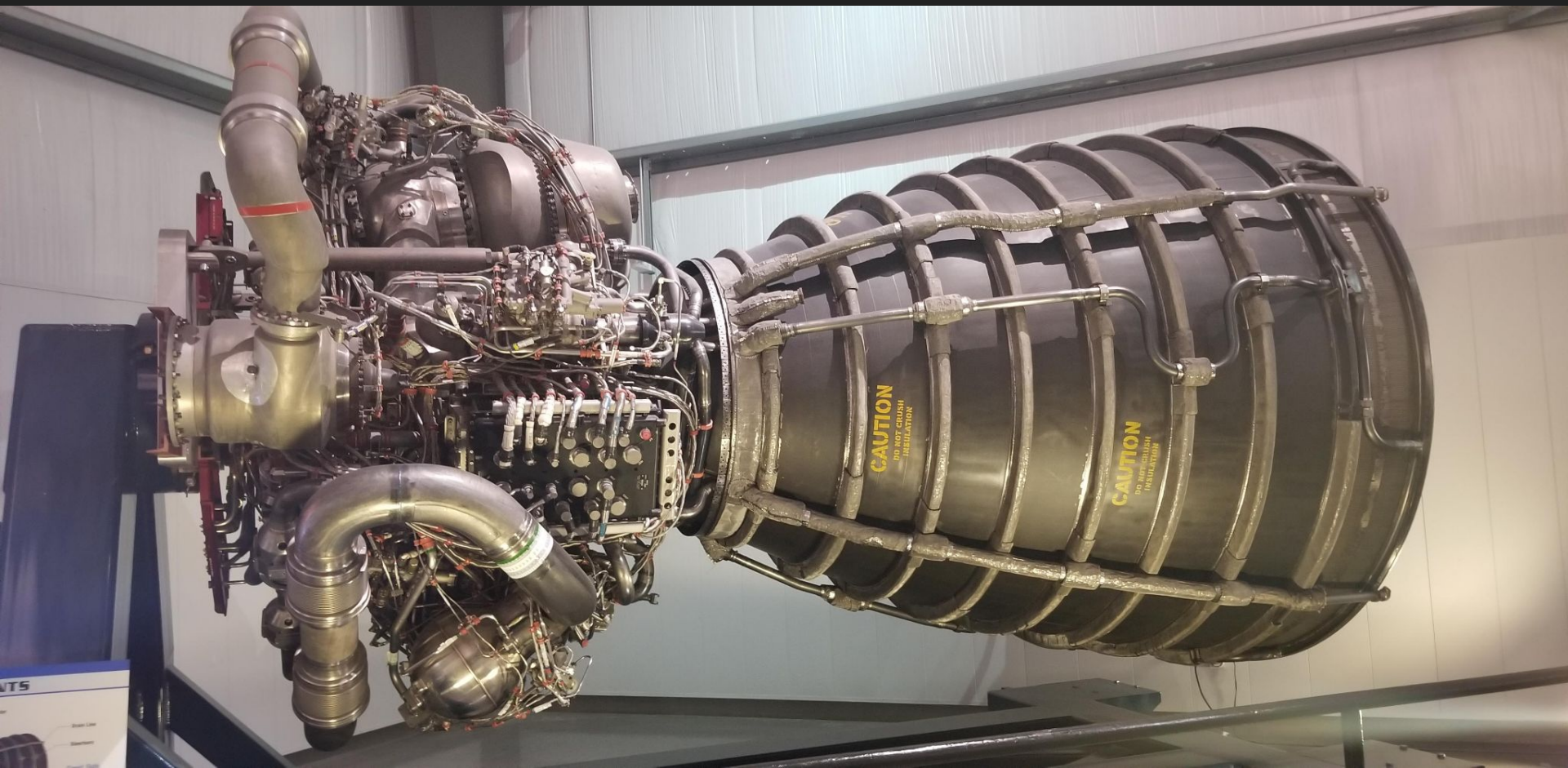


ENDEAVOUR









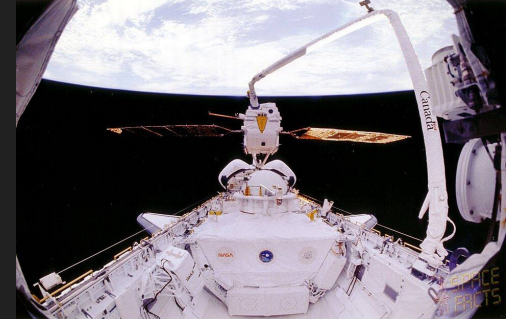
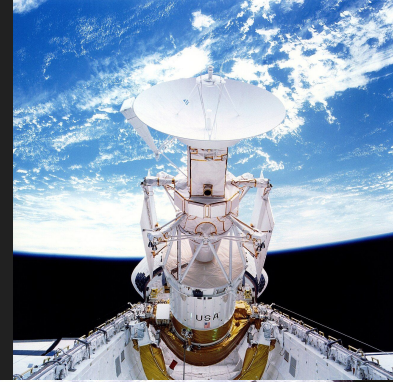
Space Shuttle Missions

- The space shuttles flew 135 missions from 1981 to 2011
- They flew for a combined 3 years, 235 days, 18 hours, 9 minutes, and 44 seconds
- 355 people from 16 countries flew on the space shuttles
- When they reached space, the space shuttles would keep orbiting around Earth just like the ISS
- Discovery had the most flights with a total of 39
- 1992-2003: we had four shuttles operating at the same time
- Sally Ride became first American woman in space in 1983 on the space shuttle (first woman was Valentina Tereshkova in 1963)



Space Shuttle Missions

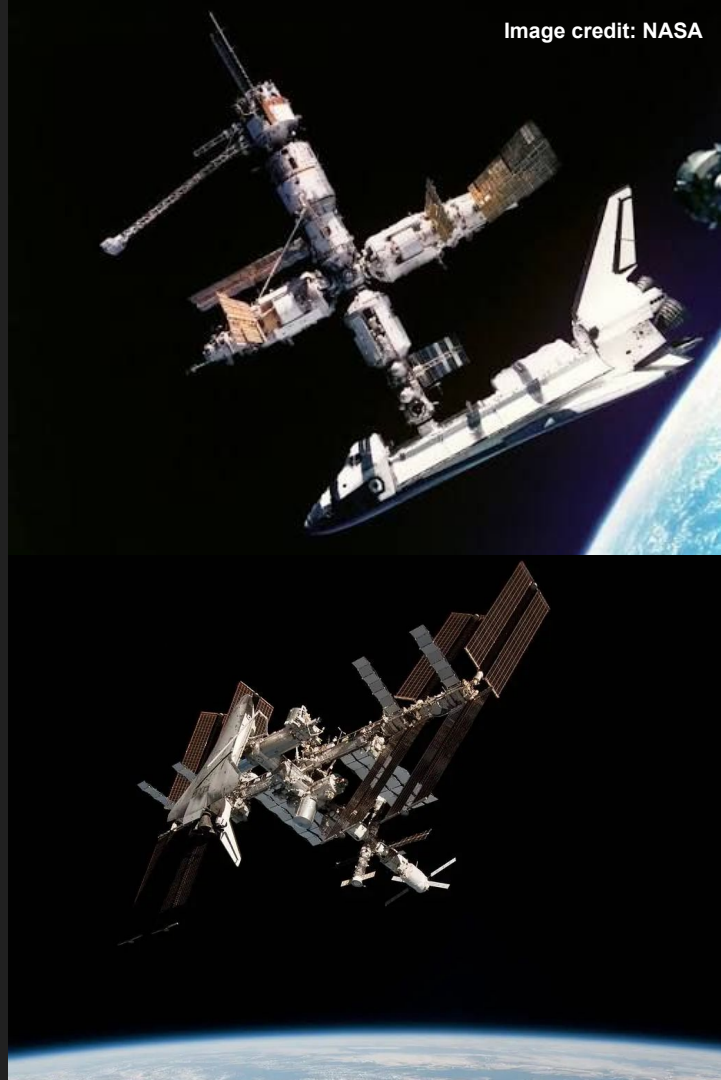
- The space shuttles could carry large payloads in the ship
- They deployed many communications satellites that are important for GPS, the internet, and phone calls
- They deployed satellites that were sent to other planets (Magellan Venus probe)
- They deployed many space telescopes for astronomy but also to study Earth's atmosphere
- They also had 10 missions for the Department of Defense (8 were classified)
→ Spy satellites obviously



Space Shuttle Missions: To the Space Stations

- Building the ISS
- 37 shuttle missions went to the ISS
- The space shuttles were incredibly important to constructing the ISS
- The ISS is a multi-module space station and the space shuttles brought many pieces up to build the ISS
- Mir (Soviet/Russian space station): docked 9 times with space shuttles

Image credit: NASA



Space Shuttle Missions: To the Space Stations

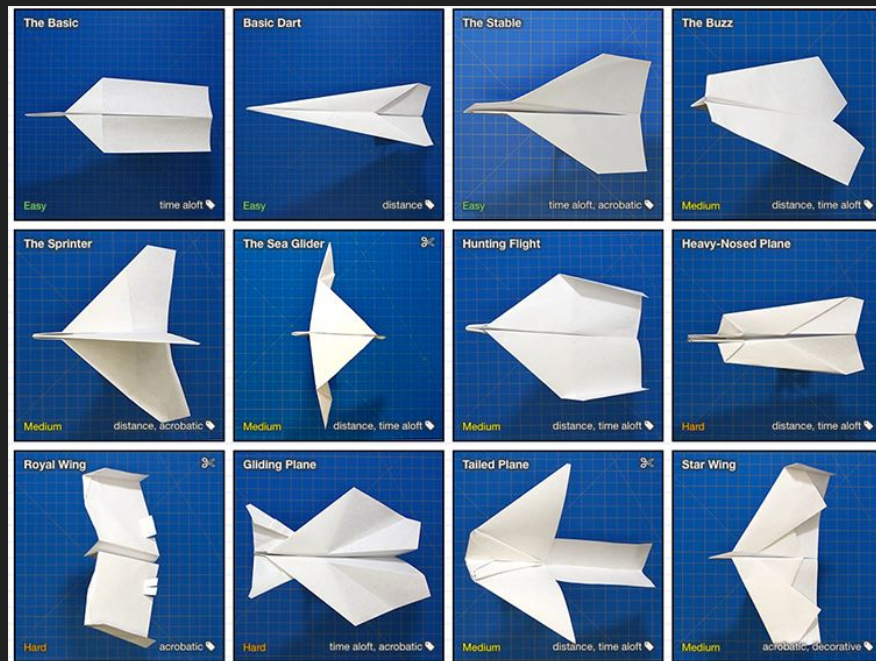
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Image credit: NASA



Today's Activity: Paper Airplane Competition

- Break up into teams
- Make your best paper airplane
- You have two tries to see how far it goes
- Can you build a paper airplane like the space shuttle?



Space Shuttle Disasters

- The Challenger (1986) and Columbia (2003) space shuttles exploded
- Challenger exploded after liftoff because the weather was too cold and parts of the rocket boosters failed at low temperatures
- Columbia exploded during reentry (landing) because damage during liftoff made the heat shield fail
 - NASA knew about the potential damage while the Columbia was in space, but did not think it was bad enough to destroy the shuttle on reentry
 - This type of damage had happened before and never resulted in an explosion
- 14 astronauts died on these missions
- There are videos, but I will not show them in class



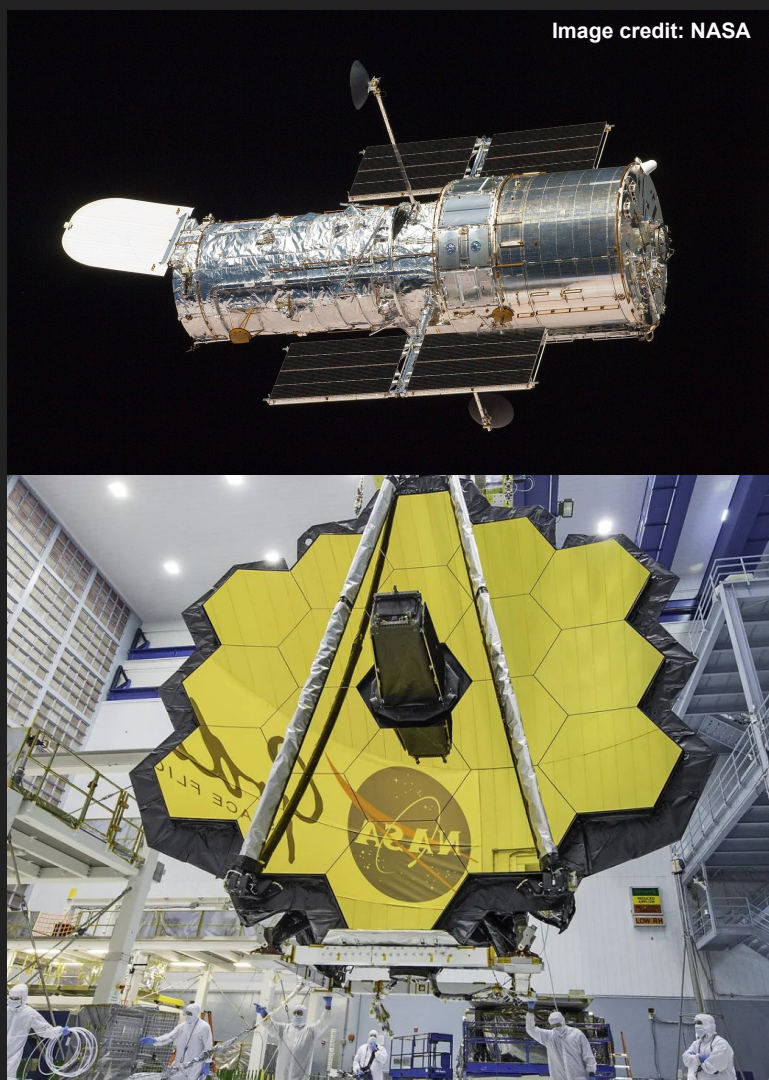
Where Can you see the Shuttles today?



Space Telescopes

- Why do we put telescopes in space?
- We can see more type of light and much clearer than on Earth
- There have been dozens of telescopes that orbit Earth and about 20 are currently operating
- Most telescopes are launched on rockets and are deployed remotely, e.g. James Webb Space Telescope

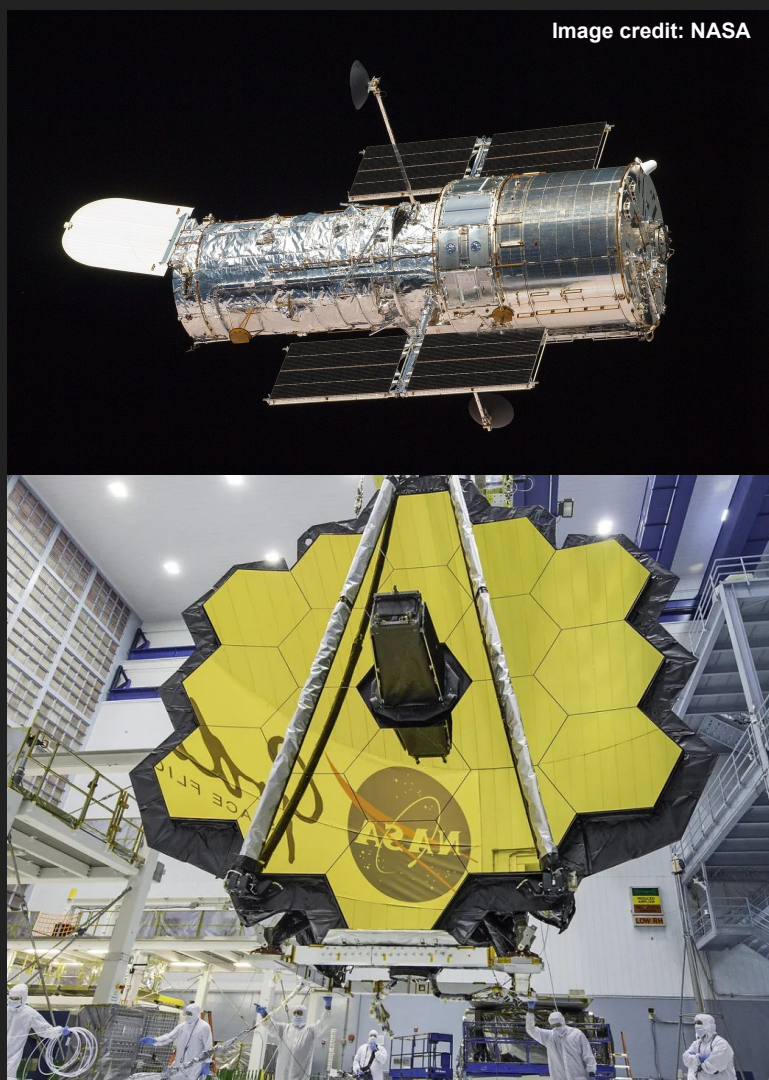
Image credit: NASA



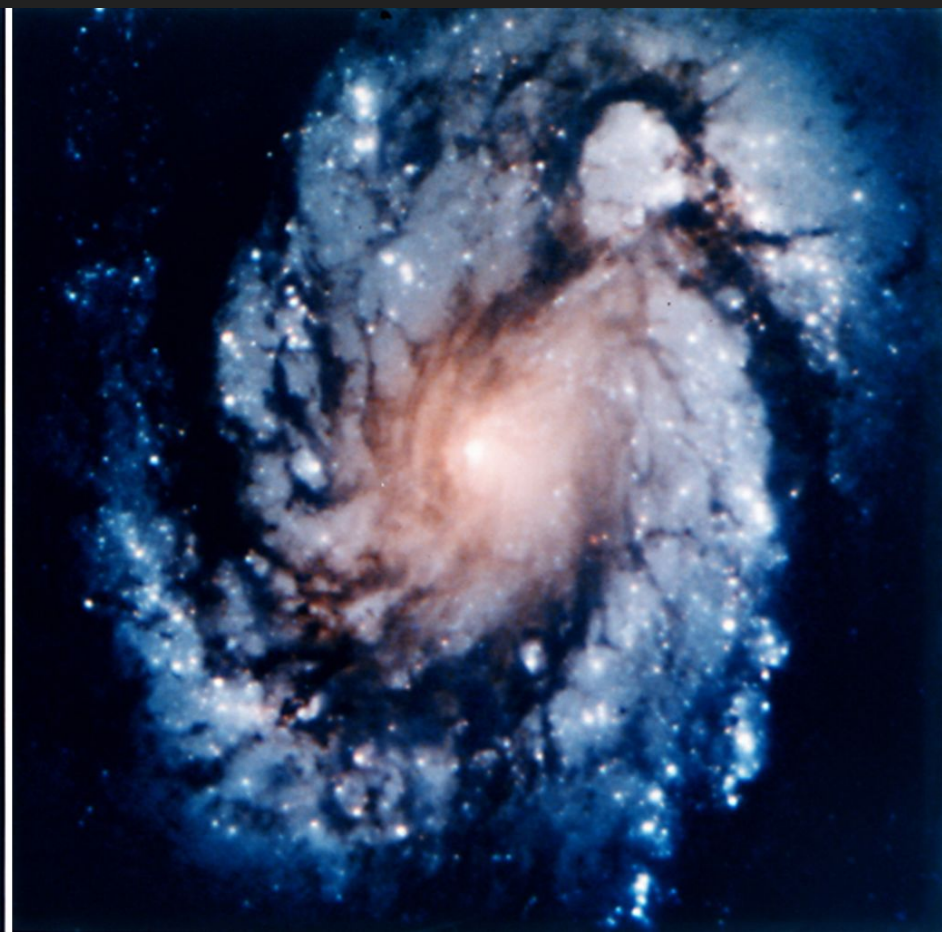
Space Telescopes

- Some have been deployed by astronauts on the Space Shuttle
- The most famous is the Hubble Space Telescope
- It orbits around Earth once every 90 minutes
- It had a major design flaw: its mirror was too flat so light was not focused correctly and images were blurry

Image credit: NASA

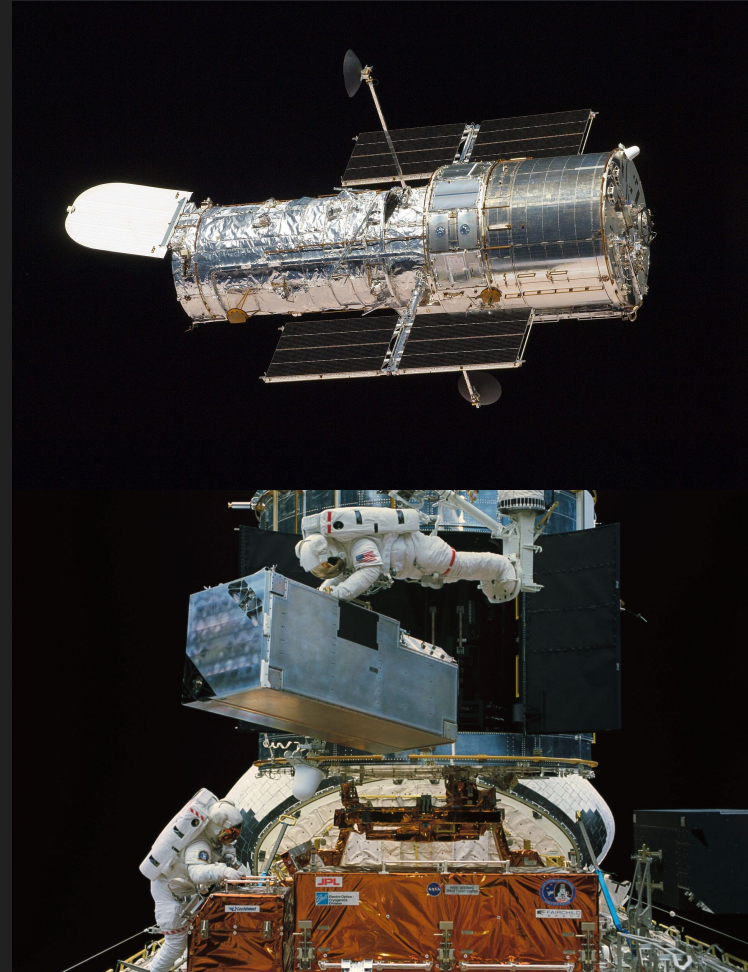






Space Telescopes

- Scientists built a machine that acts like eyeglasses to overcome the mirror error
- Because Hubble is in Earth orbit, astronauts could visit it and place new instruments inside the telescope
- Hubble has had over a dozen instruments, something unheard of for space telescopes
- Today, Hubble is one of the most accomplished telescopes ever created



A photograph of an astronaut in a white spacesuit working on the Hubble Space Telescope. The astronaut is positioned in the lower center, reaching towards a large, orange, circular component of the telescope. The background shows the complex structure of the telescope and the Earth's surface with blue oceans and white clouds. The text "SERVICING MISSION 1" is overlaid in large white letters, and "HUBBLE SPACE TELESCOPE" is overlaid in smaller white letters on a red rectangular background.

SERVICING MISSION 1

HUBBLE SPACE TELESCOPE

